



Deep cross-modal integration with hierarchical multi-head attention for big five personality prediction

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Received: 11 May 2024 / Revised: 14 June 2025 / Accepted: 25 June 2025

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Abstract

Non-intrusive personality detection identifies an individual's personality traits without directly asking them questions or conducting personality tests. Research on personality prediction is growing in various applications such as career guidance, healthcare, education, social interactions, marketing, and advertising. Presently, the methods for non-intrusive personality detection are based on multiple modalities of video, audio, and text with limited accuracy as they don't use cross-modality relations between features learned from each modality. This work explores a novel approach to learning cross-modality relations between the modalities using a cross-hierarchical multi-head attention and fusion network (CHMAFN). This architecture uses a multi-head attention mechanism to understand and capture the cross modality relations between modalities extract the relevant features at different levels of abstraction and fuse the attention to obtain the cross relation within and between the modalities. The proposed modal extracts deep learning features from multiple modalities of video, audio, and text. Then, video features are enhanced using modified DenseNet architecture, and text and audio features are enhanced using 1D CNN. Next, these features are enriched using cross-hierarchical attention and fusion networks to classify them to the big five personality traits using multivariate long short-term memory (LSTM.). The proposed network is experimented with CharLearn First Impression Dataset and the performance of the framework was found to be efficient with an accuracy of 93.97, which is 1.3% higher compared to existing works Cross Attention Transformer, MMMU-BA, Deep Multimodal Fusion, Multitask Transformer, ERINE and Cross Attention with Bilinear Transformation. The F1 score of CHMAFN increased at least by 1.32% with balanced and higher precision and recall values, which is at least 7% higher and this balance is due to the avoidance of learning bias through feature enrichment. The model CHMAFN is tested against three different datasets, VPTD (Video Dataset for Personality traits detection), UDIVA 0.5, and MATRICS (Multimodal-Task-Oriented Group Discussion), and recorded a 3 to 5% improvement in the result.

Keywords Cross-modal attention · CHMAFN · Multivariate LSTM · Multi-head attention non-intrusive

1 Introduction

Personality analysis from various modalities is growing in the field of psychology and computer science due to its potential applications in various areas like social relation analysis [1], depression analysis [2], and candidate screening for jobs [3]. The unconscious behavior of a person transcends verbal or non-verbal expressions that reflect the personality of an individual, evidence of this psychological behavior is often seen in job interviews. Personality computing involves estimating one's personality using various modalities of facial expressions, audio, speech, scene, and others. Over the years many models have been proposed for describing personality traits and the most popular among them is the Big Five personality trait. The Big Five model also called as OCEAN model, is based on the 16 Cattle theory where the human's 16 personality traits are majorly grouped into five factors that describe five core personality traits of a person like Openness: Imaginative, receptive to new experiences, and prepared to consider abstract ideas, Conscientiousness: Plan, focus on the details, set priorities, and complete critical activities on time, Extraversion: Meet and greet with a wide range of people with enthusiasm as an extrovert who enjoys engaging with them, Agreeableness: Caring with others and enjoys lending a hand to those in need and Neuroticism: Stress and worry are extremely frequent tense and go through sharp mood swings. These personality factors represent a range between two extremes of low and high [4]. The model is widely used by many researchers as it considers a wide range of cultures, demography, age, and so on. Also, it is annotated by Amazon Mechanical Truck(AMT) and made available in the dataset used in this research.

ChaLearn First Impression V2 [4] is the dataset used in this research, which includes audio, visual cues, and transcription data, which are considered the three modalities of this work. Then these modalities can be combined in different ways like, data fusion, early fusion, decision-level fusion, and hybrid fusion. Among these fusions, the early fusion will focus on improving the accuracy as it understands the insights of features used and this research uses early fusion. Current techniques are based on multiple modalities of video, speech, and text for personality prediction, and the features from various models are fused either early or late without considering the cross-correlation between the features, which introduces a learning bias, and classification accuracy is reduced. This work proposes a multi-modal deep learning technique called cross hierarchical multi-head attention and fusion network (CHMAFN) for big five personality trait classification as presented in Fig. 1. This attention network uses the features extracted from each modality to learn enriched features by exploiting the cross-reference correlation between features of each modality. The spatial and temporal correlation between the enriched features is used to classify the five traits using multivariate long short-term memory (LSTM).

The CHMAFN is a deep learning architecture to processes the information from different modalities such as audio, text, and video. Cross-attention mechanism for integrating the modalities is used to increase the accuracy. The modal extracts the dependencies within and between the modalities to understand the relationship in data. It is multilayer hierarchical where each layer is based on an attention mechanism to concentrate and use only the relevant information thus leading to improved accuracy. From the First impression video dataset, the images are extracted using Visual Geometry Group (VGG_Face model), audio extraction is through MFCC technique, and text using glove embedding, and then the extracted are sent to feature encoding. The feature encoding process is to convert the

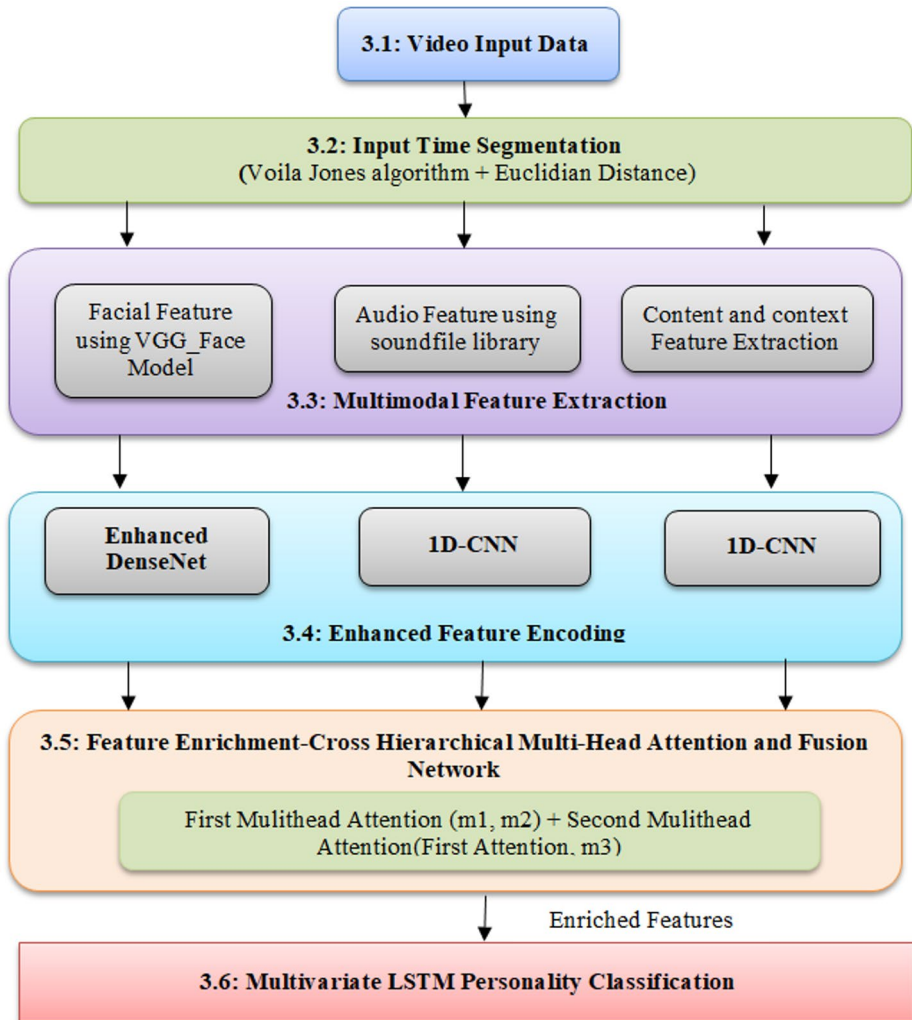


Fig. 1 The CHMAFN Architecture for prediction of big five classes

extracted features into more useful information and for encoding image features we use the enhanced densenet architecture, for text and audio feature encoding we use 1D CNN. The intention of using CNN architecture is that it learns automatically and trains the output as per the attention process at the next level. Next, the feature enrichment is done by combining the features from different modalities in a cross-modal attention mechanism. Here the cross-modal attention mechanism is based on the Multihead attention mechanism where the features from modal one say image are used to calculate attention weights for the features of model two say text, this weighted feature gives only the critical part that participates in increasing the accuracy. Then hierarchical attention mechanism to learn the local and global dependencies and multivariate LSTM extracts the time series data (spatial and time data) to classify the dominant personality from the five different traits. The discriminative ability

of enriched features in classifying Big Five personality traits is tested using entropy and clustering analysis. The following are the key contributions of this work are listed below.

1. This work proposes a novel cross-hierarchical multi-head attention and fusion network (CHMAFN), which includes cross-attention, hierarchical structure, and fusion to improve accuracy.
2. This work brings the novelty by enhancing the DenseNet architecture and using it for deep feature representation from images.
3. Usually, 1D CNN is commonly used for image processing, but in this paper, 1D CNN is specifically designed for sequential data like audio and text.
4. The paper provides a holistic approach to text extraction by considering both content and context along with sentiment lexicons called emoticons. Here domain-specific sentiment with pattern incongruity detection was used to improve the prediction accuracy.

The novelty of this research is the extended version of multi-head attention proposed in [12] for multiple modalities, where the multi-head attention is included within modalities and fusion of multi-head attention among the different modalities in a cross-hierarchical structure is implemented, which is neither discussed nor implemented before as per authors best knowledge.

The organization of the paper is as follows: Sect. 2 discusses the related work. Section 3 discusses in detail the proposed CHMAFN architecture. Section 4 discusses the result analysis, and the conclusion with future work is discussed in Sect. 5.

2 Related work

The approaches for personality prediction can be split into two categories unimodal and multimodal. Unimodal approaches use features from visual, audio and text feeds to classify personality traits. Multimodal approaches [5] use the integration of multiple feeds to classify personality. Multimodal approaches perform better compared to unimodal approaches. A survey of multimodal approaches is presented below. The work in [6], proposed a speaker embedding called i-vector to improve the prediction accuracy of the video input. Here the author combined the audio, language, and motion image differently to predict five classes of the Big Five model using a pre-trained modal. This method showed better accuracy only for traits like neuroticism and extraversion, the author considered another two datasets written in a different language to improve the score, by this conscientiousness increased when fused with motion-related features. Depression detection using a multimodal framework [7] from facial and verbal input uses the Temporal Attentive Pooling (TAP) method which involves segment-level fusion. Authors proposed a new feature technique called multimodal factorized bilinear pooling which solves the problem of overfitting. A multimodal recurrent neural network [8] used temporal information of video and text for personality prediction in contrast to transformer-based architecture. From the dataset (UDIVAv0.5), image and text are extracted and trained using LSTM [9], then using bimodal RNN, the text information is extracted and given to the BIGRU for the classification and aggregation however, the technique was only effective for the extraverted personality trait.

Deep learning architecture [10] predicts personality traits using multimodal behavioral information. The video was used to extract facial and body landmark statistical traits, text was processed to obtain sentiment features and text embeddings, and audio features as mel-frequency cepstral coefficients (MFCC). The modal independently predicts through decision fusion and the integration failed to consider the interdependencies among the several modalities. The work in [11], built a multimodal personality detection system integrating speech, text, and video. Temporal information, 200 dimension word embedding, and facial landmarks are fused to identify the personality. The author used both early and late fusion but did not take into account cross-referencing learning between the multimodalities. Transformer-based architecture [12] combines a person's facial expressions with his speech, body language, and environment to predict personality. To classify personalities, thirteen distinct behavioral traits are stored, and this is done in addition to position encoding called behavior encoding using LSTM. Here for the text analysis short memory used is not very relevant and fusion for inferring the personality was at the decision level where the cross station transformer is not considered. VGG deep learning model [13] is used to extract visual and audio features to find the person's personality from the interview videos. Feature fusion using an extreme learning machine (ELM) but cross-learning is not considered.

Using the foundations of cognitive psychology, a multimodal personality classification system is cognitively driven [14], upon viewing the image, the user provides written comments as input to predict human characteristics, including romantic, humorous, and sad, then classified using the reinforcement method on picture region and opinion words. Different transformer topologies for personality perception were studied in [15]. The author concluded from the experiment that tasks involving personality perception are better suited for linear structures. This study ignored text in favor of analyzing solely audiovisual modalities. How to integrate cross-modal learning in the presence of text was not taken into consideration in this work. Multimodal deep architecture [16] uses audio, visual, and text to classify big data personality traits. Facial appearance features from facial action units using CNN-GRU, body pose using Open Pose, voice, and transcribed speech are extracted. Feature fusion is done as early fusion, late fusion, and hybrid fusion, and personality is classified using various deep learning architectures but the cross relation between the features is not studied. Graph Neural network [17] combined face images with audio and nonverbal signals for personality prediction. The cognition of the user is represented as a graph and this graph is sent to a graph neural network for classifying personality. The personality traits are not big five personalities. Furthermore, the suggested paradigm is not universal and requires customized training for every individual.

CR-Net [18], for classifying the big five personality traits from audio, video, and text feeds. The Resnet-34 modal was used with the Bell loss function to perform the regression to predict the personality. The Chalearn First Impression dataset was used, but the cross-relation between the features was not considered. This work [19] investigated methods for eliminating bias from a multimodal personality recognition system based on deep learning. Adversarial learning and data balance were investigated. It was possible to eliminate learning bias using both methods. In terms of fairness measures, adversarial learning performs better than data balancing, although data balancing maintains or even increases modal accuracy. However, the learning was centered on individual modality and was not cross-referential. Lin et al. [20] employed multimodal information taken from speech and video to forecast a person's level of emotional expressiveness. The ability to predict emotional

expressiveness is achieved by combining facial features with language. Three techniques were tested for prediction: multilayer perceptrons, support vector regression, and elastic net. The prediction accuracy was improved by combining language behavioral information, yet the approach was only tested on smaller datasets. Deep Bimodal Regression Framework (DBR) [21] utilized audio and video to predict personality traits, The regression framework considered the audio and to forecast the final personality, considered both early and late fusion but fusion failed to take into account the cross-modal relationships. MTCNN and ResNet [22] for ambient features, VGGish CNN for audio, and n-Gram CNN for text features are used. Then these features are combined with different personality prediction models like concatenation, average method, and different attention (Interactive attention, QKV, weighted attention, and Multimodal Multi-Unit-Bidirectional Attention(MMMU-BA)). A deep multimodal fusion [23] approach predicts personality through audio, video, and text. ViT-B16 and VGG model for visual features, VGGish for audio, and Glove embedding for text. Attention-based concatenation and averaging for feature fusion were used and in which the averaging showed a better result. The study [24] uses aesthetics to predict personality traits. Advanced techniques like a swim transformer and an attention mechanism are used to achieve a strong result compared to the existing system. They used the Flickr-AES Database and PsychoFlickr database to demonstrate and prove that this outperformed the existing SOTA(state-of-the-art) methods. The approach [25] used pre-trained transformer-based models like BERT, RoBERTa, ALBERT, ELECTRA, ERINE, and XLNet in personality recognition. They developed the user-centric recommendation system in education, e-commerce, and human resources. These models captured the long-term dependencies in the hierarchical relationships for the personality classification based on the text Data. The My Personality dataset is used. The study proved that ERNIE improved the accuracy of personality prediction. Time-based modality alignment [26] using spoken word timestamps between text, audio, and visual. Using BiLSTM and self-attention to capture dependencies and key behavioral cues, then used the gradient-based domain to generalize. First self-attention between the models to focus on the modal features and then cross-modal fusion extracts the complementary relationships between modalities. Cross-modal attention with Bilinear transformation for better prediction which improved the accuracy of 78.44%.

3 The proposed CHMAFN architecture

The cross-hierarchical multi-head attention and fusion network (CHMAFN) architecture for personality classification using the ChaLearn First Impression dataset is described in this section. The multimodal inputs considered in this work are video, audio, and text, and cross-dependencies between these models are exploited using cross-modal attention. The CHMAFN architecture, shown in Fig. 1 implements the cross-modal attention process to extract only the relevant information, which is useful to learn the interdependencies between the modalities. The structure is hierarchical which represents the multiple levels, which helps in learning the different levels of abstraction by capturing high-level and fine-grained features. The attention mechanism is to focus and capture the input data and select only the relevant data that contributes for the personality prediction and ignore the rest. The CHMAFN is implemented using a deep learning algorithm such as CNN to learn the

patterns in the data. Instead of fixed time duration, the time duration for segmenting the inputs is found by marking salient regions and time segmenting the inputs. As the data pre-processing step, the input video is divided by marking the salient region, which gives the dynamic changes that occurred in the captured frames instead of dividing the video into fixed frames, this adds the dynamics and well captures the expressions created by the humans. From the obtained video segments, frames are recorded as the image modality, and audio and text data are also extracted confirming that the three different inputs extracted (visual, audio and text) are confined to the same video segment. Later separate deep models are used to process the extracted inputs from the previous steps, which results in collecting the specific features from each modality. Then in the enhanced feature encoding phase, the extracted features are encoded using enhanced Densenet for Visual features and 1DCNN for audio and text. Then the model passed to the feature enrichment stage, where the extracted features are enhanced with relevant information from the other two modalities hierarchically. The CHMAFN architecture First takes two inputs audio and visual inputs, the cross-modal attention for the audio modal uses the output from its feature encoding layer as the query vector while the output of the visual modal is the key and value vectors. Then Multihead scaled dot product attention is applied, allowing modality audio to learn cross-referenced contextual information from modality visual. By this process both the modality enhances their features through attention. Since we are dealing with multimodal inputs, this cross-modal attention is enhanced using a Hierarchical cross-reference approach, which is clearly explained in the below sections. CHMAFN has five stages: (i) Data Collection (ii) Input time segmentation (iii) Multimodal feature extraction (iv) Enhanced feature enrichment (v) Feature Enrichment- cross hierarchical multi-head attention and fusion network (vi) Multivariate LSTM Personality Classification. The video clip of the first impression dataset is collected and pre-processed in the first stage and its detailed explanation is discussed in Sect. 3.1. In the second stage, the frames from the video clips are identified and marked for segmentation and a segmented time window will be given to the next stage of prediction which is explained in Sect. 3.2. In the third stage, the image, audio, and text features are extracted using different algorithms which are explained in Sect. 3.3. After the extraction, the features are encoded with the DenseNet algorithm for image, and the 1D CNN algorithm for audio and text, which is shown in Sect. 3.4. Next, the cross-modality attention with hierarchical relation and personality classifier multivariate LSTM is clearly explained in Sect. 3.5 and 3.6. The below section explains the CHMAFN architecture.

3.1 Data collection (video input data)

First Impression V2, is the dataset used utilized in the Job Candidate Selection Competition. 10,000 video clips were considered from around 3000 HD YouTube videos to create the dataset, and the labels were created using Amazon Mechanical Turk (AMT) [4]. The dataset includes the Big Five Personality Scores in addition to voice, text, and video inputs. Extraversion, agreeableness, conscientiousness, negativity, and openness are the five classes, and their scores fall between 0 and 1. Figure 2 displays a snapshot of the dataset labeled. The professionals at Transcription Service Rev have transcribed every video clip for the text data. With a total of 435,984 words transcribed, we have 43 words per movie on average, consisting of 18 non-stop words and 14,535 unique words. These words are saved, for



Fig. 2 The video snapshot and personality traits of first impression dataset with the labels [4]

instance, as a key-value pair. The transcriptions for the video called “video_name” would be returned by running transcription[‘video_name’]. Using these videos, data is prepared for the research by extracting the frames, audio, and text data, which is then used for the prediction model.

The dataset contains public YouTube videos of individuals speaking in front of the camera, which contains demographic information of people like age, gender, and ethnic background, also contains facial expressions, language fluency, emotions, different backgrounds, lighting conditions, camera angles, clothing, speaking naturally, scripted and unscripted

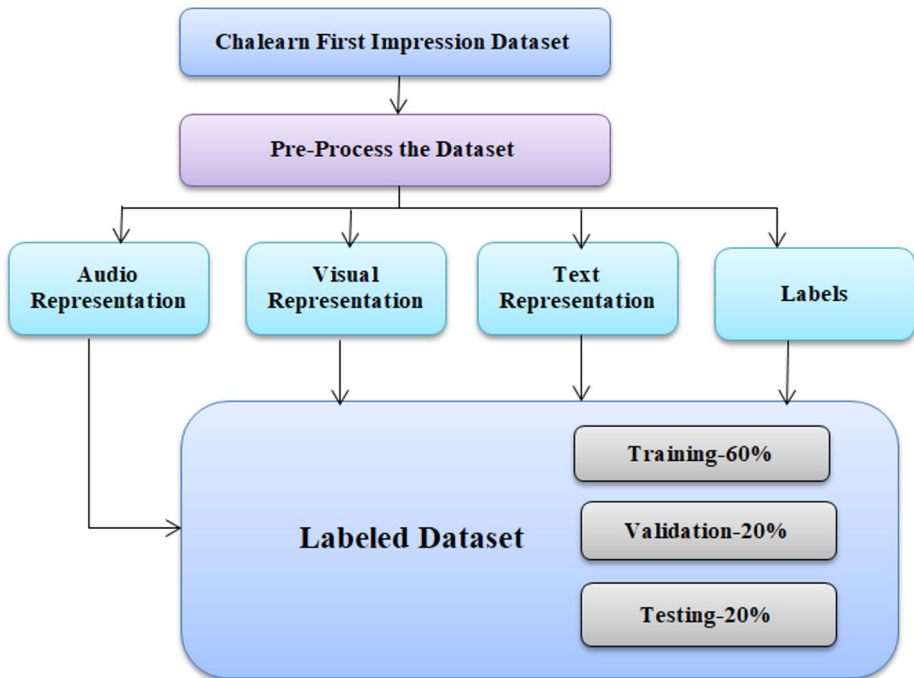


Fig. 3 Split of the dataset into training, testing, and validation-Pre Processing Stage

speech with different variation in tone, emotions with pitch. The dataset preparation is explained in Fig. 3. Where the length of train data is 60%, for validation 20%, and for test it is 20%. From 10,000 video clips, a total of 28,709 images, 1440 audio files, and 56,000 text embedding are trained. The dataset obtained contains 99 rows with 168 feature dimensions for audio and video files and for text 56,000 embedding with 32 max sequence lengths.

3.2 Input time segmentation

In this work we find out the difference in face region for the video input segmentation, to find out the difference in face region, we first detect the faces using the Voila Jones algorithm [27] from the video using the landmark feature [28] as shown in Table 1. Once the modal has identified these landmarks, we can calculate the distance or similarity between them to determine if there are any significant differences between the two faces.

Algorithm 1 extracts the anchor landmarks and mouth counter landmark features, where anchor marks are fixed from where the other points are measured. Mouth contour landmarks are specific points along the outline of the mouth, such mouth corner and the top and bottom of the upper and lower lips. The Euclidean distance between the features between frames is calculated and when the distance is greater than a threshold, then the time of the frame is marked for segmentation, which indicates the different face region.

Algorithm 1: Detection of face landmarks for input time segmentation (time frame window)

Start

Input: load video(video_path = path/video.mp4')
 A predefined threshold k

1. import cv2
2. face_cascade = cv2.CascadeClassifier(cv2.data.haar_cascades + 'haarcascade_frontalface_default.xml')
3. video_path = 'path/video.mp4' # Open a video file
4. cap = cv2.VideoCapture(video_path)
5. while True:
6. ret, frame = cap.read()
7. if not ret:
8. break
9. img = cv2.resize(frame, (224, 224)) # Preprocess the frame
10. gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY) # Convert the frame to grayscale(preprocess)
11. faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5, minSize=(30, 30))
12. for (x, y, w, h) in faces:
13. cv2.rectangle(frame, (x, y), (x+w, y+h), (255, 0, 0), 2)
14. img= cv2.imshow('Face Detection', frame) # Display the frame
15. def detect_landmarks(img):
16. landmarks = detect_landmarks_using_dlib(img)
17. return landmarks
18. prev_landmarks = None
19. threshold = k
20. landmarks = detect_landmarks(img)
21. # Calculate distance between corresponding landmarks
22. if prev_landmarks is not None:
23. for i in range(min(len(prev_landmarks), len(landmarks))):
24. dist_val = dist(prev_landmarks[i], landmarks[i])
25. if dist_val > threshold:
26. # Mark the time frame
27. print("Time frame marked") # Time frame marked for segmentation.
28. prev_landmarks = landmarks

Output: frame marked for input time segmentation.

End

Table 1 Facial features extracted using VGG_FACE models required for prediction

Name	Notation	Description
Mouth outer	f1	Distance between anchor landmarks and mouth contour landmarks
Mouth Corner	f2	Distance between anchor landmarks and mouth corner landmarks
Eye area	f3	Eye contour region area
Eyebrow Activity	f4	Distance between anchor and eyebrow landmark
Face Area	f5	Area of polygon drawn around external landmarks
Face Motion	f6	The Euclidean distance average compared to the previous frame

Algorithm 1 captures the frames from the video input and performs the face detection with landmarks for each frame, then calculates the Euclidian distance between each landmark of two frames and marks the frame for segmentation. The landmarks of the face obtained are shown in Fig. 4.

3.3 Multimodal feature extraction

3.3.1 Facial features

With the time frame, the significant frames are obtained using structural similarity. The significant frames within the time segment are the frames with significant differences compared to other frames. It is found by calculating the structural similarity of each frame to its previously found significant frame and when the structural similarity is higher than the threshold, the frame is selected as significant frame. The score ranges from -1 to 1 , where 1 represents the images perfectly matched and -1 represents the two completely dissimilar images. It is proved that the SSIM metric is the most advanced metric than other because this measure

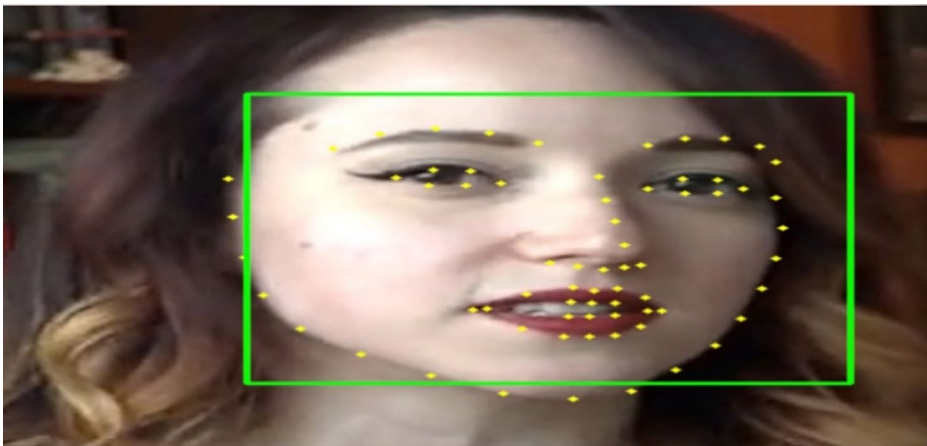


Fig. 4 Snapshot of facial image with landmarks used for image features extraction

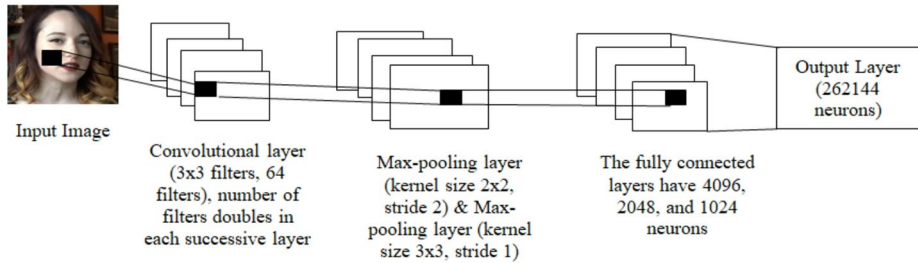


Fig. 5 The VGG_FACE_MODEL Configuration for spatiotemporal feature extraction

is based on 3 components: brightness, contrast, and structure. Also, the score is calculated pixel-wise and results in a floating point value. From each of the significant frames, faces are extracted. The faces are passed for spatio-temporal feature extraction, because they contain important visual information that can be used for various applications, and the process represents the faces in a compressed and meaningful way. The SSIM in this work is calculated using the deep neural model called the VGG Face model, the model determines whether two faces belong to the same person. It consists of 16 convolutional layers, three fully connected layers, and an output layer as softmax. The model Configuration with the last layer has 262,144 neurons, which corresponds to the number of classes for facial recognition as shown in Fig. 5. The algorithm 2 for the VGG face model is explained below.

Algorithm 2: Significant frame(Vgg_Face)

Start

Input: Load Image

1. $img1 \leftarrow load_img('path', size=224 \times 224)$
2. $img2 \leftarrow load_img('path', size=224 \times 224)$
3. $img1_ary = img_to_array(img1)$
4. $img2_ary = img_to_array(img2)$
5. # Load the VGG-Face 16 model
6. $face = VGGFace(model='vgg16', include_top=False, input_shape=(224, 224, 3))$
7. # Extract features
8. $img1_features = vgg_face_model.predict(img1)$
9. $img2_features = vgg_face_model.predict(img2)$
10. # Flatten the feature representations
11. $img1_flat = img1_features.flatten()$
12. $img2_flat = img2_features.flatten()$
13. # Calculate SSIM
14. $ssim_value = ssim(img1_flat, img2_flat, full=True)$
15. $find_correlation \leftarrow input\ img1, img2.$
16. $clusters \leftarrow Sklearn.cluster.K-means$
17. $Calculate\ Silhouette\ score \leftarrow cluster_quality(clusters)$

Output: Set of significant frames used as the training data.

End

3.3.2 Audio features

The audio samples are extracted from the time segments marked. From the sample, the following features and their descriptions are shown in Table 2. The algorithm 3 explains the extraction of audio features [29].

Algorithm 3: For audio feature extraction**Start****Input:** audio file using soundfile library**Descriptions:** The function will extract MFCC, chroma features, Mel spectrograms, and Linear Predictive Coding (LPC) features from the audio file and return a single array containing all the features.

1. 'soundfile' $\leftarrow$ read audio file
2. 'librosa' $\leftarrow$ find(MFCC, chroma, Mel Spectrogram)
3. 'LPCC' $\leftarrow$ This is the custom 'LPC' function, which extracts Linear Predictive Coding (LPC) features from the audio signal. The main functionality of LPC is to separate the speech signal from the background noise.
4. 'hstack' $\leftarrow$ This NumPy function is used to concatenate multiple arrays horizontally. The 'hstack' function concatenates the extracted features horizontally into a single array. This simplifies the feature extraction process and makes it easier to integrate with other audio-processing libraries and frameworks.

Steps:

1. Extract_audio $\leftarrow$ find(file, mfcc, chroma, mel)
2. Read AudioFile $\leftarrow$ dtype="float32"
3. X $\leftarrow$ audiofile
4. if X is chroma:
5. then: stft $\leftarrow$ find(stft(X)) # Short Time Fourier Transform
6. return(hstack(result[]))
7. if X is mfcc:
8. then: mfcc $\leftarrow$ find(librosa.mfcc)
9. return(hstack((result, mfcc)))
10. if X is lpcc:
11. lpccfe $\leftarrow$ LPC(file_name)
12. return(hstack((result, lpccfe)))
13. if X is Mel:
14. Mel $\leftarrow$ find(librosa.melspectrogram)
15. return(hstack(result, Mel))

Output: hstack of audio features.**End**

3.3.3 Text features

Texts based Personality classification is widely used [30], the texts within the time segments are extracted are subjected to feature extraction. Most of the text feature extraction models are based only on content and don't consider the context using the large language model [31, 32]. To solve this, an integrated content and context feature is proposed in this work. Texts

Table 2 Extracted audio features using sound file library

Audio Features	Description	Formulas
Linear Prediction Cepstral Coefficients (LPCC)	Captures the Spectral features like pitch, tempo, loudness, and other acoustic characteristics [29]	$LPCC[n] = c[k] + \sum_{n=0}^{N-1} x[n] \cdot x[n-k]$ LPCC[n] represents the nth LPCC coefficient C[k] denotes the kth cepstral coefficient obtained from LPC
MFCC [29]	The MFCCs capture the spectral shape of the audio signal.	$C_n = k^2 \sum_{k=1}^K \log(S[k] \cdot \cos(K\pi n(k - 0.5)))$ Where, Cn: nth Mel-frequency cepstral coefficient. K: Number of mel filters S[k]: Power spectrum of the signal at the kth frequency.
Mel Spectrogram	Visual representation of the spectral content [29]	$\text{Mel}(f) = 2595 \cdot \log_{10}\left(1 + \frac{f}{700}\right)$
Chroma	Represents the musical content information. Reveals the energy distribution [29]	$\text{chroma}_m = E(m) / \sum E(m')$ Where, E(m)- energy at m band E(m')- sum of energy of m band

within the time window are processed to extract the following content and context features. Steps for extracting text features:

1. **Separate the content and context information from the text:** done by tokenization, where the sentences are broken into tokens (words, any characters, punctuation) followed by removing punctuation and removing stop words, as the initial processing step.
2. **Content features:** These features are extracted from the distribution of words and special symbols like emoticons. Word distribution and semantic relation between words are learned using Glove embedding. Glove is a powerful word embedding algorithm. In this unsupervised technique, the word vector representation for text corpus is learned and thus Glove is called a global vector. Here the algorithm converts the word and special symbols to vectors according to the application and its training data. It is done by reducing the dimension of the co-occurrence matrix. Emoticons are special symbols that carry information about expressions like, positive, sarcastic & negative, and their frequency in the document is counted and an emoticon feature vector is constructed. The below mentioned steps are involved in textual feature embedding.
 - a Load the pre-trained glove model: In this research, we use 'glove.twitter.27B.200d.txt' as a pre-trained GloVe model with 27 billion parameters and a dimensionality of 200. Were 200-dimensional glove vectors trained on a 27-billion-word corpus.
 - b Build a Pre-trained Embedding layer: Build embedding_layer with embedding type, dimension, and length as inputs. The embedding type will consider the glove, emoji, and some random, based on the type the vectors are loaded for the representation. The embedding function for input sequence t1 is shown in Eq. 1.

$$\text{Input Sequence } (t1) = \text{Glove } (t1) + \text{Emoji } (t1) + \text{Random } (t1) \quad (1)$$

Where,

GloVe(t1)=golve{W1,W2,W3.....Wn}.

Emoji(t1)=emoji{W1,W2,W3.....Wn}.

Random(t1)=random{W1,W2,W3.....Wn}.

- c Build a Dense Neural Network (dnn_architecture): from the pre-trained embedding layer, hidden_uints, and dropout rate, a dnn_architecture. The model takes input tensor tweet_indices and passes to a pre-trained embedding layer built in the previous step to obtain the embeddings(vectors) and this is passed to the custom deep neural network architecture(LSTM), which has multiple dense layers and activation layers at the end. Here the output tensor is passed to two neuron-dense layers and an activation layer with softmax activation. This function returns the Keras model with predicted probabilities as shown in Fig. 6.

3. **Context features:** This information includes the surroundings, which will be helpful to extract the exact meaning of the word along with its relationship for example “The cat is on a mat”, the context can be the cat is in the room. Contexts can also be Sentiment information, which is a metric used to quantify the sentiment found in texts. It consists of two parts: extrinsic and intrinsic. The term “intrinsic” describes a person’s inner emotional condition. An event topic that serves as a risk indicator is referred to as extrinsic. The domain-specific

sentiment lexicon approach described is used to extract intrinsic sentiment information from text [15]. In this approach the lexicon is built, which is a list of words with corresponding sentiment scores (specific to a particular domain) for example in the Movie domain, if the customer feedback is “Good”, the sentiment score is positive and the feedback is “Bad”, then the sentiment score is negative. Here the output is a polarity score (+1/−1/0), where the +1 score is the strongly positive sentiment, the −1 score is the strongly negative sentiment and 0 is the neutral sentiment. Latent Dirichlet Allocation (LDA) is used for extrinsic sentiment analysis [15]. This approach allows extracting the sentiments at the document level, which gives the overall sentiment of the corpus using which emotions and attitudes are better understood. LDA considers the document as a mixture of topics and every topic is a word distribution each topic is added with a sentiment score (either positive/negative), and then the document sentiment score is the weighted sum of the sentiment scores of each topic. The vector score obtained for each topic score is the output. The sequence of sentence incongruity is not mandated and it represents the mismatch of the sentence and it is identified by comparing the use of verbs, pronouns, and context information. For example in the sentence “The session was good but I did not understand”, here the first part has a positive sentiment (Good), while the next part has a negative sentiment (did not). According to Camp [33] Table 3 presents the incongruity patterns, which are formed by POS (parts of Speech) tagged and its pattern count is given as an incongruity feature (s^i). The content and context features in the time window are joined to create the text feature vector. The text features are passed to 1D-CNN to get enhanced features.

3.4 Feature extraction and enhanced feature encoding

3.4.1 Enhanced densenet

In most existing works, spatio-temporal features from face regions are extracted using VGGNet. Compared to VGGNet, DenseNet learns more complex features. DenseNet is a new neural network with densely connected convolutional networks. The difference between DenseNet and Resnet is that the Resnet takes the previous output as the input for a future layer but the DenseNet takes all the previous output (concatenation) as an input for a future layer. The Resnet Model of CNN creates a vanishing gradient problem as the depth increases which reduces the performance, but the DenseNet model improves the learning and extracts the more complex features by having dense blocks, which consist of filters and maintain unique dimensions. A transition layer for batch normalization is placed between each block. Also to find the unique dimensions a down sampling is performed. DenseNet

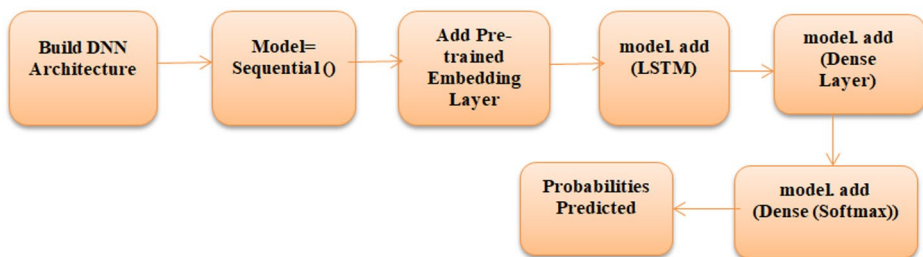


Fig. 6 Pre-trained embedding layer

becomes overfit and computational complexity is higher. This work brings the novelty by enhancing the DenseNet architecture and using it for deep feature representation from images. A modified version of the DenseNet model is used to enhance the performance, where the fully connected layer is replaced with a convolutional layer and stride layer in place of the pooling layer as shown in Fig. 7. The kernel size of 1×1 , 21 channel depth, and also an accumulation layer for the accumulation of extracted features. The modified Densenet is explained below.

- a. Replacing fully connected layers with convolutional layers: this modification reduces the number of parameters and improves the generalization in the network. We have used 1×1 convolutional layers to retain the spatial features of the video inputs and also it is an efficient feature selector that accepts variable size inputs and multiple frames. This replacement also reduced overfitting.
- b. Stride layer in place of average pooling layer: it performs the downsampling of feature maps and stores more special information than the pooling layers. Pooling is a fixed operation and non-learnable. But stride optimally learns the downsampling and preserves the spatial detail from the video frames, this inclusion captures the micro-expressions, subtle changes, and other expressions relating to personality.
- c. Accumulation layer: this layer is used in accumulating the extracted features, which help in capturing long term dependencies across the multiple inputs and also the time-stamps. From the first point we are preserving the spatial data, the intention to include the accumulation layer is to retain the temporal data. As the temporal memory unit in the attention mechanism, this maintains the temporally enriched representation. Temporal behaviors like eye movements and smiles are very important in predicting the personality.
- d. The kernel size of 1×1 , 21 channel depths: for the optimized task.

The DenseNet-121 architecture includes, Initial convolutional layer, Dense Block (B1), Transition Layer (T1), Dense Block (B2), Transition Layer(T2), Dense Block(B3), Transition Layer(T3), Dense Block(B4), Stride layer and Accumulation Layer. The input to DenseNet is the sequence of frames, which is passed to the initial convolution layer to extract the basic features. Then these features are passed to a dense block consisting of

Table 3 Text features with incongruity patterns formed using POS tags

Candidate term (Verb positive/negative)	Positive/Negative patterns	Example
Verb	Verb followed by verb	I watch a comedy movie and I laugh
Verb present participle	Verb followed by adverb	He laughed continuously
Verb gerund	Adverb and followed the verb	He laughed heartily at her joke
Verb past participle	Verb followed by a proposition	She suggested that we should go out for a movie
Verb past form	Verb followed by an adjective	He has been feeling anxious about his presentation all week
Verb present participle third person singular	Verb followed by a noun	She read a book

multiple dense units, each performing a series of operations, Batch Normalization, ReLU activation, and 3×3 convolution layer, and the output of all densenet units are concatenated. In between each dense block, a transition layer is used to reduce the spatial dimensions of the feature maps, the transition layer includes batch normalization, 1×1 convolution, and 2×2 average pooling with a stride of 2, these specification reduces the parameters size and reduces the complexity as the networks go deeper. In the next stage, all the extracted features are accumulated in the accumulation layer, these data include both the temporal and spatial data which is required in the next level.

3.4.2 1D-CNN

The power of CNN [34] learns the features automatically. The audio and text features extracted are passed to 1D-CNN for higher level representation of the audio and text features. 1D CNN is a neural network used for processing sequential data like one-dimensional signal input such as audio, text signals, and time series data. The configuration of the 1D-CNN used in this work is given in Table 4. The CNN architecture includes a large kernel size that captures the long range temporal dependencies using prosodic patterns from the audio waveform and contextual word patterns from the text feature. In the convolution operation itself stride of 2 is applied instead of pooling, this design reduces the downsampling

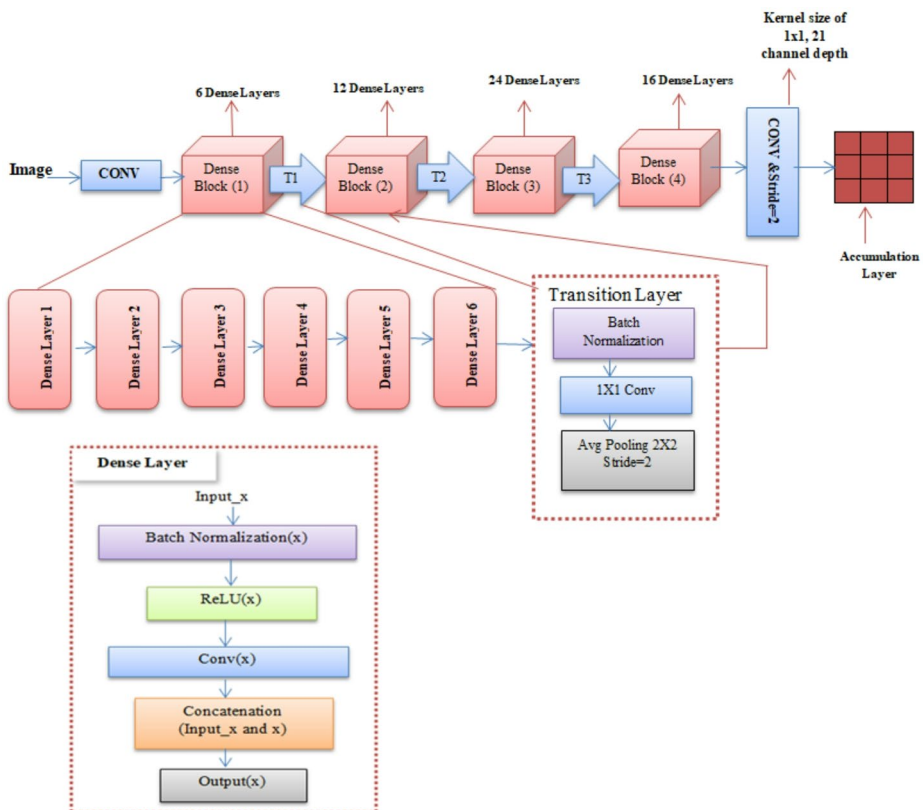


Fig. 7 Enhanced DenseNet for encoding image based on DenseNet-121 architecture

by preserving the spatial and temporal data better than the pooling. The stride also helps in aligning phoneme shifts in audio and word position in the text with no loss of data. Max pooling 1D is used in a selective manner to preserve the temporal data like, it preserves the word orderings in the text and frequency resolution intact in audio. As in the deeper layers, the filter size is reduced from 10 to 8 to reduce the modal complexity and overfitting. Flattened to the single dense layer to perform the high-level encoding of the complex sequential features, 512 neurons will balance the encoding and cost.

This work includes the construction of neural network architecture for time series classification using 1D convolution layers and max pooling. The layer descriptions are as follows:

1. 1D Convolutional layer: layer has 10 filters and 128 kernel size which applies a sliding window of size 128 to the input time series, and applies 10 different filters to extract features. Then the output is given to a ReLU activation cell. Stride is set to 2, meaning that the window moves by 2 samples at a time. The convolutional 1D layer has 8 and 128 filters and kernel size, this layer has fewer filters than the previous ones which is in the middle of the network.
2. MaxPooling 1D layer: The pool and stride size are 2 to reduce the feature maps spatial dimensions. Also, the network prevents overfitting.
3. Flatten layer. To flatten into a single vector for the next step.
4. The dense layer has 512 neurons and a ReLU activation function. This layer applies a fully connected weight matrix to the flattened input, then, applies ReLU activation for non-linearity output.

3.5 Feature enrichment- cross hierarchical multi-head attention and fusion network

The feature represented by multiple modalities is enhanced using cross-modal attention. Multi-head attention proposed in [12] is extended for multiple modalities of attention in this work as shown in Fig. 8. Say there are two modalities $\{m_1, m_2\}$. The cross-modal attention for modal m_1 takes the output of its feature encoding layer as a query vector and the output of m_2 feature encoding layers as key and value vectors. It then applies multi-head scaled dot product attention. It helps each modality to learn cross-reference information from other modalities. Finally features from the cross-modality attention of $m_1 m_2$ and are pooled and passed to prediction using a softmax classifier. This cross-modal attention proposed in [12] is enhanced using a novel hierarchical cross-reference approach for multiple modalities. In this, the cross modalities feature learned from two modalities $\{m_1, m_2\}$ is passed to the next cross-modal attention-taking output of $\{m_{2+i}\}$ the feature encoding layer for learning cross-reference information between $\{m_1, m_2\} \{m_{2+i}\}$ and. This process is repeated

Table 4 1D-CNN configuration used for encoding audio and text features extracted from the previous step

Layer	Configuration
Convolutional 1D	10*128, ReLU, Stride=2
Convolutional 1D	10*128, ReLU, Stride=2
MaxPool 1D	Size=2,Stride=2
Convolutional 1D	8*128, ReLU, Stride=2
Convolutional 1D	8*128, ReLU, Stride=2
Flatten	-
Dense	1*512,ReLU

till all N modalities are covered. The final feature is then passed for temporal personality classification. The steps with the equation are described below: In the equation $m1$ is text, $m2$ is audio and $m3$ is the image features. The detailed steps and equation are derived by considering $m1$ as the text feature, $m2$ as the audio feature, and $m3$ as the image feature. Figure 8 demonstrates the proposed work CHMAFN, basically designed to fuse and enrich the $m1, m2$, and $m3$ modalities layer by layer. From the bottom layer initial cross-modal attention between $m1$ and $m2$ features, in this block cross multi-head attention helps to attend the one modality from another, by this two modalities $m1, m2$ understands and learns the shared representations. Pooling is to reduce the dimensionality and aggregate the key features. The feature fusion layer combines the attended feature and in the Middle layer, cross-modal attention was with the fused feature and $m3$ modality, this brings the complex inter modality dependencies. In both layers, we perform multi-head attention and pooling and in the next layer final feature fusion, once all the modalities have been completed through two stages in a hierarchical fashion, we concatenate the enriched features and fuse further, which is then passed to the classifier. In summary, multi-head attention captures intra-modal dependencies in the input sequence and cross-attention aligns and contextualizes one modality's features with another pooling reduces noise and aggregates important signals and hierarchical fusion builds the features layer by layer from pairwise fusion to full multimodal enrichment. This hierarchical fusion enables fine-grained and complex modal interaction. Since we already use the encoded feature self-attention is not implemented to reduce the overhead.

4 Multi-head attention between $m1, m2$ (text, audio)

Equation 2 shows the first attention between text and audio

$$\text{Attention}_{\text{First}} = \text{MultiHeadAtt}(\text{Text}, \text{Audio}) \quad (2)$$

Q (Query), K (Key), V (Value) are defined for the text as Q_{Text} , K_{Text} , V_{Text} and for audio as Q_{Audio} , K_{Audio} , V_{Audio} .

Weight matrix for text as W_Q^{Text} , W_K^{Text} , W_V^{Text} and for audio as W_Q^{Audio} , W_K^{Audio} , W_V^{Audio} .

Now, compute the learnable projection Matrix as shown in Eqs. 3 & 4,
For text,

$$\begin{aligned} Q_{\text{Text}} &= \text{Text} \cdot W_Q^{\text{Text}} \\ K_{\text{Text}} &= \text{Text} \cdot W_K^{\text{Text}} \\ V_{\text{Text}} &= \text{Text} \cdot W_V^{\text{Text}} \end{aligned} \quad (3)$$

For audio,

$$\begin{aligned} Q_{\text{Audio}} &= \text{Audio} \cdot W_Q^{\text{Audio}} \\ K_{\text{Audio}} &= \text{Audio} \cdot W_K^{\text{Audio}} \\ V_{\text{Audio}} &= \text{Audio} \cdot W_V^{\text{Audio}} \end{aligned} \quad (4)$$

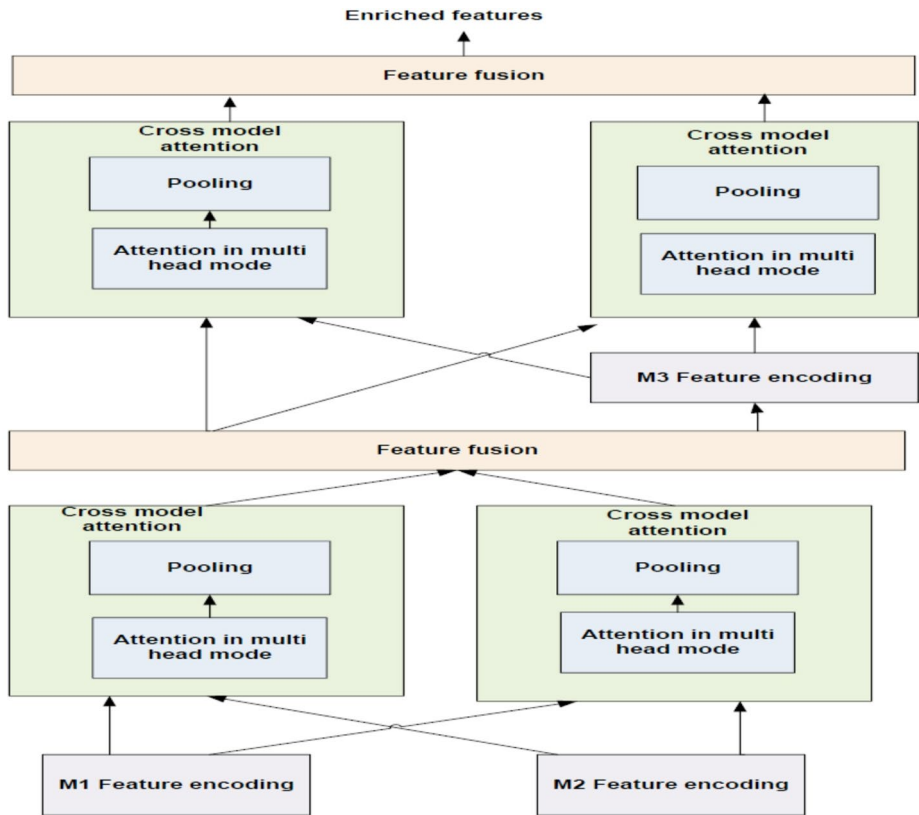


Fig. 8 Feature Enrichment with a cross attention mechanism for the three modalities image, text & audio (M1, M2 & M3)

Now, find the attention between text and audio using the dot product as in the equation.

The Multi-Head attention between text and audio is explained in Eqs.5 & 6,

$$Attn_{Text}^i = Softmax \left(\frac{Q_{Text} \cdot (K_{Audio})^T}{\sqrt{\frac{d_{model}}{H}}} \right) V_{Audio} \quad (5)$$

$$Attn_{Audio}^i = Softmax \left(\frac{Q_{Audio} \cdot (K_{Text})^T}{\sqrt{\frac{d_{model}}{H}}} \right) V_{Text} \quad (6)$$

Where,

Let H is the dimensionality.

$Attn_{Text}^i$ is the attention for the i^{th} Attention head, obtained using audio features.

$Attn_{Audio}^i$ is the attention for the i^{th} Attention head, obtained using text features.

Q_{Text} and Q_{Audio} is the Query matrix representing the text and audio feature

K_{Audio} and K_{Text} Represent the key matrix for the audio and text feature

V_{Audio} and V_{Text} Represent the value matrix for the audio and feature

d_{model} represents the dimensionality of the model's hidden states

H Represents the number of attention heads.

T denotes the matrix transpose operation.

$\sqrt{\frac{d_{model}}{H}}$ is the scaling factor, which helps stabilize the gradients during training.

5 Pooling

In Eq. 7, the attention from text and audio is pooled using concatenation, and after multi-head output, apply.

pooling for aggregating the information

$$Pool_{Text} = Concat(Attn_{Text}^1, Attn_{Text}^2, \dots, Attn_{Text}^H) \quad (7)$$

$$Pool_{Audio} = Concat Attn_{Audio}^1, Attn_{Audio}^2, \dots, Attn_{Audio}^H)$$

6 Feature fusion

The first attention is obtained using concatenating the text attention with audio attention as described in Eq. 8.

$$Attention_{First} = Concat(Attn_{Text}^1, Attn_{Text}^2, \dots, Attn_{Text}^H, Attn_{Audio}^1, Attn_{Audio}^2, \dots, Attn_{Audio}^H) \quad (8)$$

7 Second Multi-head attention between $Attention_{First}$, $m2$

The equation is similar to the first attention but uses a second set of features, which is described in Eq. 9

$$Attention_{Second} = MultiHeadAttention(Attention_{First}, Image) \quad (9)$$

Similarly, with First attention, the Multi-Head attention between first attention and the image is explained in Eqs. 10 & 11,

$$Attn_{First}^i = Softmax \left(\frac{Q_{First} \cdot (K_{Image})^T}{\sqrt{\frac{d_{model}}{H}}} \right) V_{Image} \quad (10)$$

$$Attn_{Image}^i = Softmax \left(\frac{Q_{Image} \cdot (K_{First})^T}{\sqrt{\frac{d_{model}}{H}}} \right) V_{First} \quad (11)$$

Where,

$Attn_{First}^i$ is the attention for the i^{th} Attention head, obtained using first attention (Both text and audio).

$Attn_{Image}^i$ is the attention for the i^{th} Attention head, obtained using Image features.

8 Second feature fusion

The second attention is obtained using concatenating the first attention with image attention as described in Eq. 12.

$$Attention_{Second} = Concat \left(Attn_{First}^1, Attn_{First}^2, \dots, Attn_{First}^H, Attn_{Image}^1, Attn_{Image}^2, \dots, Attn_{Image}^H \right) \quad (12)$$

9 The second attention gives the enriched features

In this step enriched feature is obtained as described in Eq. 13.

$$Enriched\ Feature = Concat(Attention_{First}, Attention_{Second}) \quad (13)$$

9.1 Multivariate LSTM personality classification

The features enriched are given to multivariate LSTM for personality classification to one of the big five classes as explained in algorithm 4. The multivariate LSTM architecture is shown in Fig. 9. Every LSTM node receives as inputs the previous hidden state and the current input vector x . With this input, the cell activation is computed as the bias (b_c) and the weighted sum of the inputs ($W_c x_t$). Using a hyperbolic tangent activation function ($\varnothing_t$), the cell activation obtained as a consequence is subsequently processed with Eq. 14 as follows:

$$c_t = \varnothing_t (W_c x_t + U_c h_{t-1} + b_c) \quad (14)$$

The cell activation h_{t-1} is the result of the preceding LSTM node in the sequence represented in Eq. 1. The hidden and input state vector weights are used as values. The gates are controlled to determine the level of activation to be remembered or forgotten. At the final state, the concealed state data is computed. How much activation must be remembered and how much must be forgotten is determined by the gates. The forget gate determines how much cell activation needs to be forgotten, whereas the input gate controls how much activation

to keep. Equation 15 computes the hidden state, the last gate is included. To produce the output vector (o_t), the final gate receives two input vectors: the forgotten vector (f_t) and the input vector (i_t).

$$\begin{aligned} f_t &= \sigma_s(W_f x_t + U_f h_{t-1} + b_f) \\ i_t &= \sigma_s(W_i x_t + U_i h_{t-1} + b_i) \\ o_t &= \sigma_s(W_o x_t + U_o h_{t-1} + b_o) \end{aligned} \quad (15)$$

Where,

$f_t \leftarrow$ A vector of forgot gate.

$i_t \leftarrow$ A vector of input gate.

$o_t \leftarrow$ A vector of output gate.

The altered original sequence $X = (X_1, X_2, \dots, X_T)$ is taken as $Z = (Z_1, Z_2, \dots, Z_T)$, input, and each Z_i represents the input embedding. T personality class observations are utilized to forecast the personality class at time $T+1$. In a regression situation, the LSTM layer's output is sent to a Softmax classifier [35]. Softmax classifier in regression settings assigns the LSTM output to one of the major five classes' potential values. The personality class prediction for the specified feature values is the output of the softmax classifier. The loss function used to train the softmax regression classifier is mentioned in Eqs. 16 & 17 below.

$$L = -[\sum_{i=1}^m \sum_{k=0}^1 1 \{y^{(i)} = k\} \log P(y^{(i)} = k | z^{(i)}; \theta)] \quad (16)$$

Where,

$$P(y^{(i)} = k | z^{(i)}; \theta) = \frac{\exp(\theta^{(k)} z^{(i)})}{\sum_{j=1}^K \exp(\theta^{(j)} z^{(i)})} \quad (17)$$

Here,

$\theta^{(1)} \theta^{(2)} \dots \theta^{(k)}$ is the model's parameter $\exp(\theta^{(k)} z^{(i)})$ and a normalized parameter for input values. The pseudo-code for personality classification is given below.

Algorithm 4: PersonalityClassify

Start
Input: Audio, Video, Text
Output: Dominant Personality(DP)
 Saltime $\leftarrow$ extract_salient_frames(video)
 Feature_vect $\leftarrow$ []
 For i=1:Saltime
 F $\leftarrow$ Extract f1 to f5 as in Table 1
 A $\leftarrow$ Extract audio features as in Table 2
 T $\leftarrow$ Extract text features as in Table 3
 Feature_vect.append({F,A,T})
 DP $\leftarrow$ LSTM.Classify(Feature_vect):
 return DP
Output Dominant Personality(DP)
End

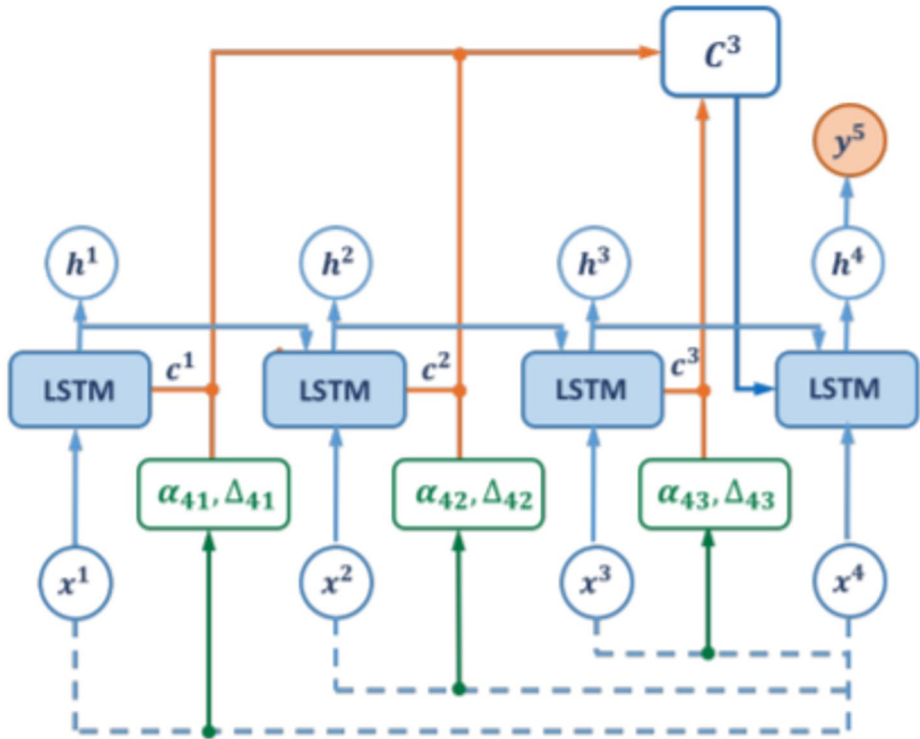


Fig. 9 Multivariate LSTM for personality classification to one of the dominant Big Five classes

10 Result analysis

The First Impressions v2 dataset [4], is used to evaluate the suggested CHMAFN performance. The dataset provides access to three modalities: text, video, and speech. The dataset videos have scores of each of the big five personalities. From the dataset of 10,000 video recordings, the training, testing, and validation split contains 6000, 2000, and, 2000 video clips, respectively. The distribution of scores for each of the personalities in the video clip is considered for the performance test which is described in the First Impression dataset [18]. An output class variable is created for each video in an experimental dataset with a dominant personality class as found from scores of the First Impressions v2 dataset. The hyperparameter for the proposed model is explained in Table 5.

The prediction accuracy for the dominant personality is measured for all the solutions as given in Table 6, and the comparison is depicted in Fig. 10. The average accuracy in predicting the dominant personality is at least 1.3% higher in the proposed solution compared to existing works. Cross-correlation learning-based fusion feature enrichment has increased the accuracy while all the existing works used feature fusion without any hierarchical cross-correlation. The performance of the proposed CHMAFN is compared against Cross Attention Transformer [12], MMMU-BA [22], Deep Multimodal Fusion [23], Multitask Transformer [24], ERINE [25], and Cross Attention with Bilinear Transformation [26]. Almost the modal performed well for all traits, but still, it is challenging for neuroticism with the low-

est performance due to emotional instability not always observable. The modal performed well for conscientiousness and extraversion which is due to better capture of energy and expressiveness.

The accuracy obtained with and without cross hierarchical multi-head attention based feature enrichment is compared and the results are given in Table 7.

Cross hierarchical multi head attention based feature enrichment has increased the accuracy by at least 9.4% compared to without feature enrichment. The obtained accuracy is measured in terms of mean absolute error between the predicted personality scores, prediction accuracy, and F1 score in terms of dominant personality. Mean Absolute Error (MAE) is calculated as mentioned in Eq. 18.

$$MAE = 1 - \frac{1}{N} \sum_{i=1}^N |y_i - \tilde{y}_i| \quad (18)$$

Where N is the test samples, y_i and $\tilde{y}_i$ is an actual and predicted personality score. The MAE is higher in the proposed solution due to joint consideration of all three modalities of text, video, and audio in the feature enrichment stage and using the enriched features for personality classification. The Fig. 11 shows the snapshot of the result obtained with precision, recall and F1 Score value.

The model is compared with the different datasets like VPTD, UDIVA 0.5, and where the modal showed 3 to 5% improvement in the result for CHMAFN as shown in Table 8.

10.1 Computational cost

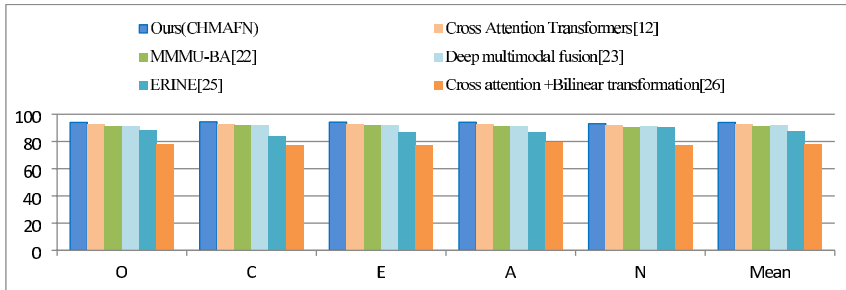
The proposed modal was trained with the NVIDIA A100 GPU(40GB), 32GB RAM, and a Tensor of 312 cores. The computational cost analysis is provided in Table 9. The table depicts the total parameters, training time, and memory used. The model took 84 s per video and 40 min for 10,000 video clips. DenseNet and LSTM for image and textual features took the most resource intensive in terms of both training time and memory usage. The GPU has 312TFLOPs(TeraFlops) for FP16(Half Precision) improved the speed with a batch size of 32 and with this optimization, the time was reduced and executed within 18 min. The model can be improved with a system RAM of 128GB and a processor to TPU(Tensor Processing Unit) to deploy easily in the real world application.

Table 5 Hyperparameters of the proposed method CHMAFN

Hyperparameters	Values
Epochs	50
Hidden Size	512
Number of Attention head	8
Dimensional Keys	64
Batch Size	32
Activation Function	Relu, Softmax
Learning Rate	5e-5
Dropout Rate	0.3
Optimizer	Adam
Loss function	MSE

Table 6 Prediction accuracy of the proposed method CHMAFN

Method	O	C	E	A	N	Mean
Ours(CHMAFN)	94.01	94.45	94.22	94.15	93.05	93.976
Cross Attention Transformers [12]	92.91	92.58	92.72	92.88	92.1	92.638
MMMU-BA [22]	91.08	91.99	91.72	91.27	90.48	91.308
Deep multimodal fusion [23]	91.5	92.27	91.81	91.44	91.52	91.708
ERINE [25]	88.09	84.11	86.53	87.16	90.56	87.29
Cross attention + Bilinear transformation [26]	78.08	77.26	77.54	79.33	77.42	77.926

**Fig. 10** Comparison of prediction accuracy for the CHMAFN model**Table 7** Comparison with feature enrichment and without feature enrichment

Personality	CHMAFN with feature enrichment	CHMAFN without feature enrichment
Openness	94.01	83.44
Conscientiousness	94.45	84.88
Extraversion	94.22	85.04
Agreeableness	94.15	86.92
Neuroticism	93.05	82.56
Average	93.976	84.568

10.2 Ablation study

The ablation study proves the efficiency of our approach CHMAFN, which is based on feature enrichment through a hierarchical attention mechanism. This complete model includes DenseNet, IDCNN, Multihead Cross modality attention, and BiLSTM fusion. Hierarchical cross-attention fusion helps in capturing both local and global temporal relationships, which significantly improves the performance by 93.976 which is 9.4% more compared to the CHMAFN without feature enrichment as shown in Table 7 where there is an improvement with all the five classes as shown. The model without feature enrichment means it does not include the hierarchical Multihead cross attention fusion resulting in an accuracy of 84.568. The simple baseline model includes DenseNet, CNN, and BiLSTM without attention shows an accuracy of 80.44, and only with DenseNet, the model recorded an accuracy of 77.06. In the feature enrichment, the inclusion of enhanced DenseNet reads the more complex relationship between the modal and boosts the performance due to its ability to capture fine-

Personality wise results				
	precision	recall	f1-score	support
O	0.72	1.00	0.84	1930
C	1.00	0.93	0.96	4347
A	1.00	0.92	0.96	3007
N	1.00	0.93	0.96	1783
E	1.00	0.93	0.97	1336
accuracy			0.94	12403
macro avg	0.94	0.94	0.94	12403
weighted avg	0.96	0.94	0.94	12403

MSE= 1.725469644440861
 Correlation coefficeint= 0.9049588226229909
 Accuracy with cross model attention = 93.97726356526647
 Accuracy without cross model attention= 84.57953720873982
 Correlation coefficeint= 0.9049588226229909

Fig. 11 Snapshot of the result obtained for the CHMAFN model

Table 8 Different dataset comparison

Dataset	O	C	E	A	N
First Impression(ours)	94.01	94.45	94.22	94.15	93.05
VPTD [36]	82	86.5	85.6	83.2	80.11
UDIVA 0.5 [8]	85.0	89.8	99.3	69.0	98.12
MATRICES [6]	59.0	68.0	75	67	71.0

grained facial features. The below Table 10 gives the ablation analysis for the components used in the architecture.

11 Conclusion

This work proposed a novel cross-hierarchical multi-head attention and fusion network (CHMAFN), which includes cross-attention, hierarchical structure, and fusion to improve accuracy. The proposed architecture includes enhanced DenseNet model for image, 1D CNN for text and audio extraction. The text extraction considered both content and context along with sentiment lexicons called emoticons to improve the prediction accuracy. Big Five personality classes are predicted using deep learning features extracted from multi-modal data such as video, audio, and text. In this work first, the features of image, audio, and text are extracted separately using VGG_Face, MFCC technique, and Glove word embedding methods. Then the extracted features are encoded using Enhanced DenseNet for image and 1D CNN for audio and text. Next, features are enriched using a cross-hierarchical multi-head attention and fusion network. Features extracted from each modality are enriched with cross-modal attention to get features with higher personality discrimina-

Table 9 Computational cost analysis of modals

Modals	Parameters	Training Time	Memory
Visual	134,268,170	35 min	6.2 GB
Audio	8,097,234	25 min	2GB
Text	11,302,652	32 min	2GB
CHMAFN	164,168,056	40 min	10.2GB

Table 10 Ablation study of proposed model

Personality	Average Accuracy
DenseNet and concatenation	77.06
CHMAFN without attention	80.44
CHMAFN without feature enrichment	84.568
CHMAFN with feature enrichment	93.976

tion ability. With consideration for both spatial and temporal characteristics, the proposed solution increased the accuracy by at least 1.3% compared to existing works like Cross Attention Transformer, MMMU-BA, Deep Multimodal Fusion, Multitask Transformer, ERINE, and Cross Attention with Bilinear Transformation. The feature enrichment process proposed in this work is scalable and it has contributed 9% to accuracy improvement. The balance between precision and recall is at least 7% higher in CHMAFN than in the existing works. The enrichment process of CHMAFN is highly scalable to extend with other modalities and is also suitable for large datasets. This model also worked well with other three datasets like, VPTD (Video Dataset for Personality traits detection), UDIVA 0.5, and MATRICS (Multimodal-Task-Oriented Group Discussion), and recorded a 3 to 5% improvement in the result. Unlike earlier methods the proposed modal maintains above 93% accuracy across all traits, indicating balanced generalization. In the future, this modal can be used with complex transformer-based architecture and can use pre-trained models for the feature extractions and prediction task which explores the transfer learning architecture. Incorporate explainability methods to improve the interpretability by visualization of weights, and may be used for real-time implementation, especially in the medical field, personalized content recommendations, and mental health assessments with lightweight optimization. To improve the accuracy of neuroticism, physiological cues can be included as a modality. The dataset contains only the pairwise comparisons and may not capture the full complexity as it is a short video, also reflects gender biases and culture so future work could explore data bias by domain adaption and improve generalization across diverse populations.

Acknowledgements Revathi S, and Dr. Nagaraja G S, (IEEE Senior Member) thank the Dept. of CSE, R V College of Engineering for their support, and Dr Farooque Azam (SMIEEE) thanks the School of Computer Science and Engineering, REVA University for facility and support towards this research.

Author contributions Revathi S: Conceptualization; Investigation; Methodology; Validation, Farooque Azam: Software; Writing – review& editing, G. S. Nagaraja: Project administration; Supervision.

Funding No funding was received for this research.
Declarations.

Data availability The dataset is freely available through the provided link <https://chalearnlap.cvc.uab.cat/dataset/24/data/41/description/>.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval Not Applicable.

Consent to publish I hereby give my consent for the publication of my manuscript in Multimedia Tools and Applications, and confirm that it has not been published elsewhere or under consideration for publication.

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